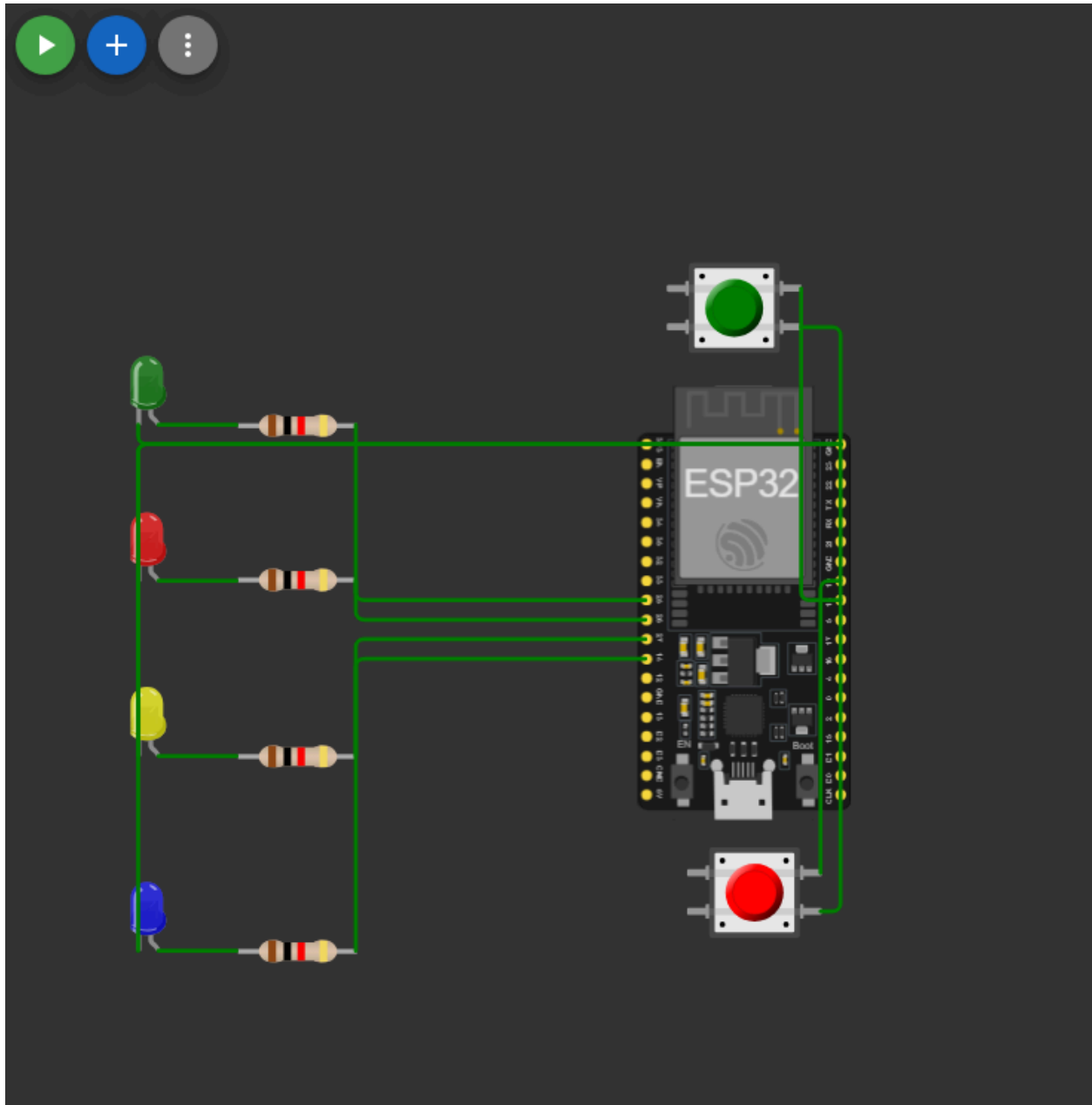


Smart Garage Gate – Future Mall

Task 5 – Screenshots

1. Wokwi Circuit

ESP32 Smart Garage Circuit



The circuit contains:

- ESP32
- 2 Push Buttons
- 4 LEDs
- 4 Resistors

2. MicroPython Code

Smart Garage Control Code

```
from machine import Pin
from time import sleep

# -----
# Pins
# -----

entry_button = Pin(18, Pin.IN, Pin.PULL_UP)
exit_button = Pin(19, Pin.IN, Pin.PULL_UP)

green_led = Pin(25, Pin.OUT)
red_led = Pin(26, Pin.OUT)
yellow_led = Pin(27, Pin.OUT)
blue_led = Pin(14, Pin.OUT)

# -----
# Garage settings
# -----

car_count = 0
MAX_CARS = 15

# -----
# Update garage LEDs
# -----

def update_garage_status():
    if car_count < MAX_CARS:
        green_led.on()
        red_led.off()
    else:
        green_led.off()
        red_led.on()
```

```

# -----
# Start status
# -----

update_garage_status()

print("Smart Garage Started")
print("Cars:", car_count)


# -----
# Main Loop
# -----

while True:

    # Car entering
    if entry_button.value() == 0:

        if car_count < MAX_CARS:
            car_count += 1

            print("Car Entered")
            print("Cars:", car_count)

            # Yellow LED blink
            yellow_led.on()
            sleep(0.5)
            yellow_led.off()

            update_garage_status()

        # Wait until button is released
        while entry_button.value() == 0:
            sleep(0.05)

```

```

52     while True:
55         if entry_button.value() == 0:
69             # Wait until button is released
70             while entry_button.value() == 0:
71                 sleep(0.05)
72             # Car exiting
73             if exit_button.value() == 0:
74                 if car_count > 0:
75                     car_count -= 1
76                     print("Car Exited")
77                     print("Cars:", car_count)
78                     # Blue LED blink
79                     blue_led.on()
80                     sleep(0.5)
81                     blue_led.off()
82                     update_garage_status()
83                     # Wait until button is released
84                     while exit_button.value() == 0:
85                         sleep(0.05)
86                 sleep(0.05)

```

The program controls the garage car count and the four LEDs according to the garage status.